



ZOTAC GeForce RTX 3060 Ti Twin Edge OC Graphics Card

Sweet spot got a little sour

Competition in the graphics card space has been quite intense over the last year since both NVIDIA and AMD have been pushing out cards at all price points and AMD's latest cards have been really threatening NVIDIA's dominance in the high-end segment. Unfortunately, both NVIDIA and AMD high-end cards are hard to come by so the advantage goes to whichever company has higher stock in the market. We've got an 8 GB NVIDIA card in a price segment which is expected to see competing AMD cards with much higher memory capacities.

The ZOTAC GeForce RTX 3060 Ti Twin Edge OC comes in a pretty small package featuring their twin blower cooling design and is as powerful as the previous gen RTX 2080 Super, if not better. Let's see how it holds up.

Build

The ZOTAC GeForce RTX 3060 Ti Twin Edge OC has 4864 shader units, 80 ROPs and is clocked at 1410 MHz. The video buffer is modest with 8 GB of GDDR6 memory when compared to the 24 GB of VRAM that the RTX 3090 cards had. The recently announced RTX 3060 12 GB of GDDR6 memory. That's because the RTX 3060 was announced after the RX 6000 series from AMD which have three cards at the moment and they all have 16 GB of video buffer. So NVIDIA's upping the game with the new cards but that also throws doubt on the RTX 3060 Ti. So we might see high memory variants of the RTX 3060 Ti but given the memory configuration, simply throwing more memory at the GPU isn't worth it since it might go unused.

Coming to the card's physical attributes, it comes with an HDMI 2.1 port, three DisplayPort 1.4a connec-



Price
₹44,500

tors, and does away with the USB Type-C connectors that the RTX 20-series graphics cards had. It runs off a standard 8-PIN PCIe power connector and the cooler is designed for a TDP of 200W. NVIDIA and ZOTAC recommend a 650 W power supply for the card. It comes with a backplate which is a good thing to have but rarely seen in the 60-class cards. But ever since the RTX 20-cards launched, we haven't seen the 60-class cards sell at the 20-25K price segment. So all these value additions are expected. Lastly, for the cooling, the card uses ZOTAC's twin-blower design which they call IceStorm 2.0.

Performance

Starting off with the synthetic benchmarks, we see that the ZOTAC GeForce RTX 3060 Ti Twin Edge OC scores about 7400 in 3DMark FireStrike Ultra. This puts it ahead of the RTX 2080 Super by about 550 points which is quite a significant lead. In Basemark, the RTX 3060 Ti scores more than the RTX 2080 Super in the OpenGL and DirectX 12 benchmarks but falls a little short in the Vulkan test. For Ray Tracing performance in 3DMark Port Royale the difference is within decimals and in certain runs, the RTX 2080 Super does score more but for the

vast majority of runs, the new 3060 Ti is better.

Coming to gaming, we ended up scoring 159 FPS in Battlefield V, 171 FPS in Death Stranding, 91 FPS in Metro Exodus, 86 FPS in Red Dead Redemption II, 147 FPS in Shadow of the Tomb Raider and 168 FPS in The Witcher

III. The RTX 2080 Super had scored 146 FPS in Battlefield V, 165 FPS in Death Stranding, 85 FPS in Metro Exodus, 75 FPS in Red Dead Redemption II, 121 FPS in Shadow of the Tomb Raider and 160 FPS in The

Witcher III. The above scores are at 1080p. At 4K, the gap between the two cards comes down to just 3-5 FPS with some games only allowing about a 1 FPS delineation. Effectively, the RTX 3060 Ti is better than the RTX 2080 Super at a much lower price.

Verdict

The ZOTAC GeForce RTX 3060 Ti Twin Edge OC Graphics Card is a wonderful addition to the new RTX 30-series albeit the memory is a bit low considering the competition. It provides performance that is equivalent to the RTX 2080 Super which costs 1.5x what the 3060 Ti does. If you've been holding out for an upgrade to your 900-series or 10-series card, the RTX 3060 Ti is a great starting point.

—Mithun Mohandas

SPECIFICATIONS

CUDA CORES: 4864 | TENSOR CORES: 152 | RT CORES: 38
| LITHOGRAPHY: 8nm Samsung | TDP: 200W | MEMORY:
8GB GDDR6 | MEMORY BANDWIDTH: 448 GB/s.

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PERFORMANCE.....81
VALUE.....75